United States Patent	12417367
Kind Code	B2
Date of Patent	September 16, 2025
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Product information systems and methods

Abstract

A product information system and method include a housing, and a communication device coupled to the housing. The communication device is configured to provide information regarding one or more products in response to being engaged by a device of an individual.

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Appl. No.: **18/347807**

Filed: **July 06, 2023**

Prior Publication Data

Document Identifier	Publication Date
US 20240020505 A1	Jan. 18, 2024

Related U.S. Application Data

us-provisional-application US 63389962 20220718
us-provisional-application US 63410307 20220927
us-provisional-application US 63420786 20221031
us-provisional-application US 63387103 20221213

Publication Classification

Int. Cl.: **G06K19/07** (20060101); **G06K19/08** (20060101); **G06Q30/06** (20230101)

U.S. Cl.:

CPC **G06K19/0727** (20130101); **G06K19/083** (20130101); **G06Q30/06** (20130101);

Field of Classification Search

CPC: G06K (19/0723); G06K (19/07703)

USPC: 235/492

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application relates to and claims priority benefits from U.S. Provisional Application No. 63/389,962, entitled “Securing Assembly Having Communication Device,” filed Jul. 18, 2022, which is hereby incorporated by reference in its entirety. (2) This application also relates to and claims priority benefits from U.S. Provisional Application No. 63/410,307, entitled “Securing Assembly,” filed Sep. 27, 2022, which is hereby incorporated by reference in its entirety. (3) This application also relates to and claims priority benefits from U.S. Provisional Application No. 63/420,786, entitled “Securing Assembly Having Communication Device,” filed Oct. 31, 2022, which is hereby incorporated by reference in its entirety. (4) This application relates to and claims priority benefits from U.S. Provisional Application No. 63/387,103, entitled “Securing Assembly Including a Suction Cup and a Coupling Layer Secured to the Suction Cup,” filed Dec. 13, 2022, which is hereby incorporated by reference in its entirety.

FIELD OF THE DISCLOSURE
(1) Examples of the present disclosure generally relate to product information systems and methods, such as can provide information about one or more products in response to being engaged by a device of an individual.
BACKGROUND OF THE DISCLOSURE
(2) Various commercial enterprises offer goods for sale. Many establishments offer a large number products for sale. Separate and distinct signs may be used to provide point of sale and product information for certain products. However, such signs occupy space, which may otherwise be used for products. Further, such signs may not provide sufficient information regarding a product. For example, the signs may not be large enough to provide sufficient details and information about a product.
SUMMARY OF THE DISCLOSURE
(3) A need exists for system and a method that allow an individual to quickly and easily determine information regarding a product, service, and/or the like.
(4) With that need in mind, certain examples of the present disclosure provide a product information system including a housing, and a communication device coupled to the housing. The communication device is configured to provide information regarding one or more products in response to being engaged by a device of an individual.
(5) The housing can include a rear panel secured to a front panel.
(6) In at least one example, the housing includes a clip that removably retains a card. The card retains the communication device.
(7) In at least one example, the clip includes a tap indicia. The tap indicia provides an area where the individual can tap the device to engage the communication device.
(8) The card can include a plurality of defined slots configured to receive and retain reciprocal protrusions of the clip.
(9) The card can includes one or both of a graphics or text regarding the one or more products.
(10) In at least one example, the communication device is secured within folded portions of the card.
(11) The product information system can also include a securing assembly coupled to the housing.
(12) As an example, the securing assembly includes a moveable joint that moveably couples the housing to a mount. The moveable joint can be a ball and socket joint.
(13) The securing assembly can include a flexible layer of nanotechnology gel.
(14) In at least one example, the securing assembly comprises a suction cup assembly. The suction cup assembly can include a flexible layer of nanotechnology gel.

(15) In at least one example, the securing assembly includes a stand.
(16) The communication device can be secured to the securing assembly.
(17) The securing assembly can include a suction cup and a suction securing nut. The communication device can be disposed between and within one or both of the suction cup and the suction securing nut. The securing assembly can also include an insert mount that fits within or around a stem of the suction cup. The communication device can be mounted on the insert mount.
(18) The communication device can be secured to the suction cup. Optionally, the communication device can be secured to the suction securing nut.
(19) In at least one example, the communication device includes a near field communication chip.
(20) In at least one example, the housing includes a front ring removably secured to a rear ring. A central passage is formed through the ring. The communication device is aligned with the central passage.
(21) In at least one example, the housing includes a front ring secured to a rear retainer. The rear retainer secures to a rail by a retaining clip and a tap screw.
(22) Certain examples of the present disclosure provide a product information method including coupling a communication device to a housing; and providing, by the communication device, information regarding one or more products in response to the communication device being engaged by a device of an individual.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) FIG. **1** illustrates a front isometric view of a product information system, according to an example of the present disclosure.
(2) FIG. **2** illustrates a front isometric view of the product information system receiving a smart card, according to an example of the present disclosure.
(3) FIG. **3** illustrates a rear isometric view of the product information system, according to an example of the present disclosure.
(4) FIG. **4** illustrates a rear isometric view of a front panel of a housing, according to an example of the present disclosure.
(5) FIG. **5** illustrates a front isometric exploded view of the product information system, according to an example of the present disclosure.
(6) FIG. **6** illustrates a front view of a card and a communication device, according to an example of the present disclosure.
(7) FIG. **7** illustrates a front view of the card securing the communication device, according to an example of the present disclosure.
(8) FIG. **8** illustrates a front isometric view of the card being coupled to the housing of the product information system, according to an example of the present disclosure.
(9) FIG. **9** illustrates a front isometric view of the product information system having the card, according to an example of the present disclosure.
(10) FIG. **10** illustrates a front isometric exploded view of a product information system, according to an example of the present disclosure.
(11) FIG. **11** illustrates an isometric view of a product information system, according to an example of the present disclosure.
(12) FIG. **12** illustrate a front view of the product information system of FIG. **11**.
(13) FIG. **13** illustrates a top view of the product information system of FIG. **11**.
(14) FIG. **14** illustrates a bottom view of the product information system of FIG. **11**.
(15) FIG. **15** illustrates a lateral view of the product information system of FIG. **11**.
(16) FIG. **16** illustrates an isometric front view of a housing, according to an example of the present

disclosure.
(17) FIG. **17** illustrates a front view of the housing of FIG. **16**.
(18) FIG. **18** illustrates a rear view of the housing of FIG. **16**.
(19) FIG. **19** illustrates a lateral view of the housing of FIG. **16**.
(20) FIG. **20** illustrates an isometric exploded view of a product information system, according to an example of the present disclosure.
(21) FIG. **21** illustrates an isometric exploded view of a product information system, according to an example of the present disclosure.
(22) FIG. **22** illustrates an isometric front view of the product information system of FIG. **21**.
(23) FIG. **23** illustrates a front view of the product information system of FIG. **21**.
(24) FIG. **24** illustrates a top view of the product information system of FIG. **21**.
(25) FIG. **25** illustrates a bottom view of the product information system of FIG. **21**.
(26) FIG. **26** illustrates a lateral view of the product information system of FIG. **21**.
(27) FIG. **27** illustrates an isometric exploded view of a securing assembly, according to an example of the present disclosure.
(28) FIG. **28** illustrates an isometric view of the securing assembly having a communication device mounted on an insert mount, according to an example of the present disclosure.
(29) FIG. **29** illustrates an isometric view of the securing assembly having the communication device mounted on the insert mount, which is disposed within a cavity of a stem of a suction cup, according to an example of the present disclosure.
(30) FIG. **30** illustrates an isometric view of the securing assembly, according to an example of the present disclosure.
(31) FIG. **31** illustrates an isometric exploded view of a securing assembly, according to an example of the present disclosure.
(32) FIG. **32** illustrates an isometric view of the securing assembly having a communication device mounted on an insert mount, according to an example of the present disclosure.
(33) FIG. **33** illustrates an isometric view of the securing assembly having the communication device mounted on the insert mount, which is disposed within a cavity of a stem of a suction cup, according to an example of the present disclosure.
(34) FIG. **34** illustrates an isometric view of the securing assembly, according to an example of the present disclosure.
(35) FIG. **35** illustrates an isometric view of a securing assembly having a suction cup separated from a suction securing nut, according to an example of the present disclosure.
(36) FIG. **36** illustrates an isometric view of a securing assembly having a suction cup separated from a suction securing nut, according to an example of the present disclosure.
(37) FIG. **37** illustrates an isometric rear view of a coupler, according to an example of the present disclosure.
(38) FIG. **38** illustrates a rear view of the coupler of FIG. **37**.
(39) FIG. **39** illustrates a lateral view of the coupler of FIG. **37**.
(40) FIG. **40** illustrates a front view of the coupler of FIG. **37**.

DETAILED DESCRIPTION OF THE DISCLOSURE

(41) The foregoing summary, as well as the following detailed description of certain examples will be better understood when read in conjunction with the appended drawings. As used herein, an element or step recited in the singular and preceded by the word “a” or “an” should be understood as not necessarily excluding the plural of the elements or steps. Further, references to “one example” are not intended to be interpreted as excluding the existence of additional examples that also incorporate the recited features. Moreover, unless explicitly stated to the contrary, examples “comprising” or “having” an element or a plurality of elements having a particular condition can include additional elements not having that condition.
(42) Examples of the present disclosure provide product information systems and methods. The

systems and methods include a housing, and a communication device coupled to the housing. The communication device can be a near field communication tag. The communication device can be contained on or within a card, and thereby provide a smart card.

(43) FIG. **1** illustrates a front isometric view of a product information system **100**, according to an example of the present disclosure. The product information system **100** includes a housing **102**, which can include a rear panel **104** secured to a front panel **105**. Optionally, the housing **102** can include a single, unitary panel. The housing **102** includes a base **106**, which can include a flat, planar surface **108**. The base **106** can be a front surface of the front panel **105**, for example. A bottom ledge **110** of the rear panel **104** extends from a lower edge **112**. The bottom ledge **110** is configured to snapably secure to a reciprocal recess of the front panel **105**.

(44) An upper clip **114** extends downwardly from an upper edge **116** of the housing **102** over an upper portion of the surface **108**. The upper clip **114** is separated from the surface **108** by a gap. The upper clip **114** includes a tap indicia **118**. The tap indicia **118** is an area where an individual can tap a smart device (such as a smart phone, tablet, or the like) to receive information regarding a product, as described herein. The tap indicia **118** provides a graphic, text, and/or the like that indicates where an individual should move a handheld smart device (for example, by tapping, waving, or otherwise moving in close proximity, such as within 10 centimeters or less) to receive information regarding product(s).

(45) The product information system **100** also includes a securing assembly **120** coupled to the housing **102**. In at least one example, the housing **102** forms at least a part of the securing assembly **120**. In at least one example, the securing assembly **120** includes a moveable joint **122** that moveably couples the housing **102** to a mount **124**. As an example, the joint **122** can be a ball and socket joint. As another example, the joint **122** can be an articulating arm. As another example, the joint **122** can be a fixed, non-movable connection.

(46) FIG. **2** illustrates a front isometric view of the product information system **100** receiving a card **130**, according to an example of the present disclosure. Referring to FIGS. **1** and **2**, the card **130** can be a planar sheet that is configured to be retained on the base **106** by the upper clip **114**. As shown, the card **130** can include a plurality of defined slots **132** that are configured to receive and retain reciprocal protrusions (such as tabs, barbs, nubs, posts, and/or the like) extending from a lower surface of the upper clip **114** to secure the card **130** in a stable position.

(47) In at least one example, the card **130** can include graphics, texts, and/or the like that provides information for a product. For example, the card **130** can include an advertisement for a product. The card **130** can include a communication device, such as a near field communication chip, as described herein. In this manner, the card **130** can be a smart card. Optionally, the card **130** may not include a communication device. Instead, a communication device can be separately secured to the housing **102**, for example.

(48) FIG. **3** illustrates a rear isometric view of the product information system **100**, according to an example of the present disclosure. In at least one example, the mount **124** includes a flexible layer **125** of nanotechnology gel, which is configured to secure the product information system **100** to a surface. For example, the mount **124** is configured to removably secure to a front surface of a door of a refrigerated compartment. As another example, the mount **124** can removably secure to a surface of a wall, a table, a countertop, a desk, and/or the like. Instead of the flexible layer **125** of nanotechnology gel, the mount **124** can be or otherwise include a suction cup assembly. As another example, the mount **124** can include the suction cup assembly having a flexible layer of nanotechnology gel.

(49) FIG. **4** illustrates a rear isometric view of the front panel **105** of the housing **102**, according to an example of the present disclosure. A rear surface of the clip **114** can include an area **134** that is configured to receive and retain a communication device **136**, such as a chip, radio frequency identification (RFID) tag, and/or the like. In at least one example, the communication device **136** is a near field communication (NFC) chip decal that is configured to couple to the rear surface **115** of

the clip **114**. The communication device **136** can be directly secured to the area **134**. As another example, the communication device **136** is retained on and/or within the card **130**, as described herein.
(50) FIG. **5** illustrates a front isometric exploded view of the product information system **100**. As shown, the front panel **105** can snapably secure to the rear panel **104**, such as through one or more deflectable tabs **138** that secure into reciprocal slots **140**. A ball **127** extending from a front surface of the mount **124** can be moveably retained within a reciprocal bearing **123** of the joint **122**. The joint **122** can include a cap **129** surrounding the bearing **123**. The cap **129** can be configured to threadably secured to a reciprocal structure on a rear surface of the rear panel **104**.
(51) FIG. **6** illustrates a front view of the card **130** and the communication device **136**, according to an example of the present disclosure. As shown, the card **130** can include a base sheet **150** and a retainer sheet **152** extending upwardly from the base sheet **150**. The slots **132** provide openings sized and shaped to align with protrusions (such as tabs) of the clip **114** (shown in FIGS. **1** and **2**, for example). The communication device **136**, such as an NFC chip decal, is secured to a defined location **154** on the card **130**. The defined location **154** is configured to be underneath the tap indicia **118** (shown in FIG. **1**, for example) by way of the protrusions being retained by overlapping slots **132**.
(52) FIG. **7** illustrates a front view of the card **130** securing the communication device **136**, according to an example of the present disclosure. As described herein, the communication device **136** can be retained within folded portions of the card **130**. For example, referring to FIGS. **6** and **7**, the retainer sheet **152** is folded over the base sheet **150** to securely sandwich the communication device **136** therebetween. After the retainer sheet **152** is folded over the base sheet **150**, the slots **132** of each are aligned, and configured to retain protrusions of the clip **114**. Optionally, the card **130** may not be configured to have foldable portions. Instead, the communication device **136** can be configured to adhere to a surface of the card **130** without any portions being folded over to secure the communication device **136** in place.
(53) FIG. **8** illustrates a front isometric view of the card **130** being coupled to the housing **102** of the product information system **100**, according to an example of the present disclosure. Referring to FIGS. **6-8**, the card **130** includes the communication device **136**. The card **130** is then inserted upwardly in the direction of arrow B so that the slots **132** engage the protrusions of the clip **114**, thereby securing the card **130** to the housing **102**. The card **130** is disposed behind the tap indicia **118**.
(54) FIG. **9** illustrates a front isometric view of the product information system **100** having the card **130**, according to an example of the present disclosure. As shown, the card **130** may be secured to the housing **102** by the clip **114**, and may extend below a lower edge of the housing **102**.
(55) Referring to FIGS. **1-9**, the card **130** having the communication device **136** provides a smart card. The communication device **136** is configured to provide product information in response to an individual placing a smart device in proximity to the tap indicia **118**.
(56) The product information system **100** can be used in conjunction with a product holder system. As another example, the product information system **100** can be separate from a product holder system.
(57) The communication device **136** can include a control unit, one or more processors, and/or the like. For example, the communication device **136** device can be a near field communication (NFC) tag. As another example, the communication device **136** can be a radio frequency identification (RFID) tag. The communication device can be on or within the card **130**. As another example, the card **130** itself is or otherwise provides the communication device.
(58) In operation, an individual positions a handheld device, such as a smart phone, tablet, or the like, in close proximity (such within 10 centimeters or less, and/or by tapping with the handheld device) to the tap indicia **118**. In response, the communication device **136** one or both of outputs information to the handheld device (such as information regarding a product, service,

advertisement, promotion, and/or the like), and/or automatically directs the handheld device to an information source (such as a website) that provides the information.
(59) As described herein, the product information system **100** includes the housing **102**, which retains the card **130**, and can attach to multiple display locations in retail environments. The card **130** having the communication device **136** is configured to provide near field communication (NFC) to mobile phones and devices at the point of display while shoppers are in the process of shopping. NFC chips can be attached to the housing **102** directly, or attached to or embedded within the card **130**, which are inserted into and/or onto the housing **102**.
(60) When individuals place a smart device (such as a mobile smart phone) within ˜1 inch of a noted tap location (for example, the tap indicia **118**) identified on the holder, the smart device recognizes the communication device **136**, such as an NFC chip, which triggers the smart device to connect to a URL address programmed into the NFC chip.
(61) The housing **102** having the clip **114** can be used either by itself or attached to the securing assembly **120**. The clip **114** allows different cards to be selectively inserted, and locked in place. The communication device **136** can be either directly applied to the clip **114** or to the card **130** depending on a desired usage.
(62) In at least one example, the securing assembly **120** includes a housing connected to a ball and socket-type mount. The ball and socket mount allows the device to be rotated and located into multiple positions as needed to face the shopper in the most optimal position. Optionally, the securing assembly can include a suction cup. As another example, the securing assembly may not include a ball and socket.
(63) In at least one example, the securing assembly **120** can include a suction cup that couples to a suction securing nut (or knob) and/or a securing mount, which can be integrally formed with a bracket, as described in U.S. Pat. No. 10,104,986, entitled “Systems and Methods for Securing and Displaying Products,” which is hereby incorporated by reference in its entirety. Optionally, the securing assembly **120** can include locking rings, or other such features that separately couple to the bracket. As another example, the securing assembly **120** can be configured as described in U.S. Pat. No. 10,393,168, entitled “Securing Assembly,” which is hereby incorporated by reference in its entirety. As another example, the securing assembly **120** can include a suction cup permanently secured to a bracket. For example, a suction cup can extend from a portion of the bracket. In at least one other example, instead of a suction cup, the securing assembly **120** can includes a flange or other such structure coated with a nanotechnology gel, which is similar to an adhesive, but can be removed and reused.
(64) FIG. **10** illustrates a front isometric exploded view of a product information system **100**, according to an example of the present disclosure. In this example, the housing **102** includes a front ring **200** and a rear ring **202**. The front ring **200** is configured to snapably secure to the rear ring **202**. A central passage **204** is formed through the front ring **200**.
(65) The communication device **136** is configured to be aligned (such as substantially axially aligned—for example, within +/−5 degrees) with the central passage **204**. For example, the communication device **136** can be secured to the card **130** within the central passage **204**. As another example, the communication device **136** can be retained within the central passage **204** between the front ring **200** and the rear ring **202** without a separate card.
(66) The front ring **200** includes protrusions **206**, such as fins, tabs, or the like, that are configured to be retained within reciprocal openings **208** of the card **130**. The communication device **136** can include the tap indicia, and is retained on the card **130** by the housing **102**. For example, an interior rim **220** of the front ring **200** may sandwich an outer rim **222** of the communication device **136** against a front surface of the card **130**. Optionally, the communication device **136** can be adhesively secured to an area **223** of the card **130**, and the front ring **200** and the rear ring **202** can then be snapably secured together and positioned on the card **130**.
(67) The card **130** can be folded over itself to securely contain the communication device **136**

between folded portions. Optionally, the card **130** may not be configured to fold over a portion of the communication device **136**. Instead, the communication device **136** can be secured to the area **223**, either directly, and/or by the housing **102**.

(68) As shown, the product information system **100** may not include a separate securing assembly. Instead, the rear ring **202** may include a coupler, such as a nanotechnology gel. As another example, the card **130** can include a strap, wire, string, or the like that is configured to allow the product information system **100** to hang from a hook, for example, As another example, the card **130** can include a coupler, such as a nanotechnology gel. As another example, the housing **102** may be configured to be retained by a stand.

(69) FIG. **11** illustrates an isometric view of a product information system **100**, according to an example of the present disclosure. FIG. **12** illustrate a front view of the product information system **100** of FIG. **11**. FIG. **13** illustrates a top view of the product information system **100** of FIG. **11**. FIG. **14** illustrates a bottom view of the product information system **100** of FIG. **11**. FIG. **15** illustrates a lateral view of the product information system **100** of FIG. **11**.

(70) Referring to FIGS. **11-15**, the communication device **136** can be retained by the housing **102**, such as any of those described herein, without a separate card. Optionally, the communication device **136** can be secured to the card, which can then be secured to the housing **102**, as shown and described herein. Further, the housing **102** is secured to a securing assembly **120**, which can be a stand **300**. The stand **300** includes a base **302** that is configured to be supported on a surface, such as a countertop, table, desk, or the like. An extension wall **304** outwardly extends from the base **302** to outwardly extend the communication device **136**. The extension wall **304** can be angled to set the housing **102** and the communication device **136** retained by the housing **102** as an angle θ with respect to the base **302**. A bottom surface **306** can include a nanotechnology gel **308**, as described herein.

(71) FIG. **16** illustrates an isometric front view of a housing **102**, according to an example of the present disclosure. FIG. **17** illustrates a front view of the housing **102** of FIG. **16**. FIG. **18** illustrates a rear view of the housing **102** of FIG. **16**. FIG. **19** illustrates a lateral view of the housing **102** of FIG. **16**.

(72) Referring to FIGS. **16-19**, the housing **102** can be used to securely retain a communication device. The housing **102** can be used with any of the examples shown and described herein. The housing **102** can be coupled to a securing assembly. Optionally, the housing **102** can be used without a securing assembly. The housing **102** can retain a communication device by itself, or in conjunction with a card, as described herein.

(73) FIG. **20** illustrates an isometric exploded view of a product information system **100**, according to an example of the present disclosure. Referring to FIGS. **11-20**, the communication device **136** can be secured to the area **223** of the card **130**. The housing **102** retains the card **130**. The stand **300** supports the card **130** and the communication device **136** in an upward position, such as when the stand **300** is supported on a surface.

(74) FIG. **21** illustrates an isometric exploded view of a product information system **100**, according to an example of the present disclosure. FIG. **22** illustrates an isometric front view of the product information system **100** of FIG. **21**. FIG. **23** illustrates a front view of the product information system **100** of FIG. **21**. FIG. **24** illustrates a top view of the product information system **100** of FIG. **21**. Figure illustrates a bottom view of the product information system **100** of FIG. **21**. FIG. **26** illustrates a lateral view of the product information system **100** of FIG. **21**.

(75) Referring to FIGS. **21-26**, the housing **102** can include the front ring **200** secured to a rear retainer **400** that is configured to clip to a rail **402** by a retaining clip **404** and a tap screw **406**. The rail **402** can be a bracket, beam, or the like. In at least one example, the rail **402** is a price sign holder. In this example, the securing assembly **120** includes the retaining clip **404** and the tap screw **406**, which are configured to secure the housing **102** to the rail **402**. In this example, the product information system **100** may not include a card. Optionally, the product information system **100**

may include a card.

(76) FIG. **27** illustrates an isometric exploded view of a securing assembly **120**, according to an example of the present disclosure. FIG. **28** illustrates an isometric view of the securing assembly **120** having a communication device **136** mounted on an insert mount **508**, according to an example of the present disclosure.

(77) The securing assembly **120** can be used with respect to any of the examples of the present disclosure. For example, the securing assembly **120** shown in FIGS. **27** and **28** can be used in place of the securing assembly **120** shown in FIGS. **1-3**.

(78) Referring to FIGS. **27** and **28**, the securing assembly **120** can include a suction cup **502** that couples to a suction securing nut **504** (or knob) and/or a securing mount, which can be integrally formed with a bracket, as described in U.S. Pat. No. 10,104,986, entitled “Systems and Methods for Securing and Displaying Products.” Optionally, the securing assembly **120** can include locking rings, or other such features that separately couple to the bracket. As another example, the securing assembly **120** can be configured as described in U.S. Pat. No. 10,393,168, entitled “Securing Assembly.” As another example, the securing assembly **120** can include a suction cup permanently secured to a bracket. For example, a suction cup can extend from a portion of the bracket. In at least one other example, instead of a suction cup, the securing assembly **120** can include a flange or other such structure coated with a nanotechnology gel, which is similar to an adhesive, but can be removed and reused.

(79) In at least one example, the securing assembly **120** includes the communication device **136**, which can be disposed between and within one or both of the suction cup **502** and the suction securing nut **504**. As noted, the communication device **136** can be a near field communication tag or chip, an RFID tag or chip, and/or the like. In at least one example, the securing assembly **120** also includes the insert mount or capsule **508** that is configured to fit within and/or around a stem **510** of the suction cup **502**. In at least one example, the communication device **136** mounts on the insert mount **508**.

(80) FIG. **29** illustrates an isometric view of the securing assembly **120** having the communication device **136** mounted on the insert mount **508**, which is disposed within a cavity of the stem **510** of the suction cup **502**. FIG. **30** illustrates an isometric view of the securing assembly **120**. As shown in FIG. **30**, the suction securing nut **504** couples to the suction cup **502**, such as via rotational threadable engagement. As shown in FIGS. **27-30**, the communication device **136** is within the securing assembly **120** at a distal end of the stem **510** of the suction cup **502** proximate to an end of the suction securing nut **504**.

(81) FIG. **31** illustrates an isometric exploded view of a securing assembly **120**, according to an example of the present disclosure. FIG. **32** illustrates an isometric view of the securing assembly **120** having the communication device **136** mounted on the insert mount **508**. FIG. **33** illustrates an isometric view of the securing assembly **120** having the communication device **136** mounted on the insert mount **508**, which is disposed within a cavity of the stem **510** of the suction cup **502**. FIG. **34** illustrates an isometric view of the securing assembly **120**. As shown in FIGS. **31-34**, the communication device **136** can be disposed within the cavity of the stem **510** proximate to a base of the suction cup **502**. The securing assembly **120** shown and described with respect to FIGS. **31-34** can be used with any of the examples of the present disclosure described herein.

(82) FIG. **35** illustrates an isometric view of the securing assembly **120** having the suction cup **502** separated from the suction securing nut **504**, according to an example of the present disclosure. As shown in FIG. **35**, the communication device **136** can be secured to the suction securing nut **504**. For example, the communication device **136** can be secured to an outer surface of the suction securing nut **504**, whether directly, or indirectly through an intermediate coupling mount. Optionally, the communication device **136** can be secured to an interior surface of the suction securing nut **504**. The securing assembly **120** shown and described with respect to FIG. **35** can be used with any of the examples of the present disclosure described herein.

(83) FIG. **36** illustrates an isometric view of the securing assembly **120** having the suction cup **502** separated from the suction securing nut **504**, according to an example of the present disclosure. As shown in FIG. **36**, the communication device **136** can be secured to the suction cup **502**. For example, the communication device **136** can be secured to an outer surface of the suction cup **502**, whether directly, or indirectly through an intermediate coupling mount. Optionally, the communication device **136** can be secured to an interior surface of the suction cup **502**. The securing assembly **120** shown and described with respect to FIG. **36** can be used with any of the examples of the present disclosure described herein.
(84) FIG. **37** illustrates an isometric rear view of a coupler **602**, according to an example of the present disclosure. FIG. **38** illustrates a rear view of the coupler **602** of FIG. **37**. FIG. **39** illustrates a lateral view of the coupler **602** of FIG. **37**. FIG. **40** illustrates a front view of the coupler **602** of FIG. **37**.
(85) Referring to FIGS. **37-40**, the coupler **602** includes a stem **610** extending from a base **612**. As shown, the base **612** can be a circular disk. The stem **610** can be coaxially aligned with the base **612**.
(86) In at least one example, the coupler **602** is used with the securing assembly **120** shown and described with respect to FIGS. **27-36**. In particular, in at least one example, the coupler **602** can be used in place of the suction cup **602**. As noted above, instead of a suction cup, the securing assembly **120** can include a flange or other such structure coated with a nanotechnology gel **620**, which is similar to an adhesive, but can be removed and reused. The coupler **602** can be used with the securing assembly **120**, or various other brackets, devices, assemblies, and the like.
(87) The coupler **602** further includes a coupling layer **603** extending from a rear surface of the base **612**. The coupling layer **603** can be a film, coating, adhesive panel, and/or the like. In at least one example, the coupling layer **603** is formed of a nanotechnology gel. The coupling layer **603** can be removable from the base **612**. Optionally, the coupling layer **603** can be permanently secured to the base **612**. The coupling layer **603** is configured to adhere to a surface, such as a glass panel. As such, the coupler **602** is used to secure the securing assembly **120** to a structure, such as a glass door, for example.
(88) In at least one example, the stem **610** and the base **612** can be formed of a polymer. In at least one example, the stem **610** and the base **612** can be formed of polycarbonate. In at least one example, the coupling layer **603** is formed of a polyurethane adhesive, which is configured to secure the coupler **602** to a structure. As such, the coupler **600** can be used in place of the suction cup **602** to removably secure the securing assembly **120** to a structure.
(89) In at least one other example, the coupling layer **603** is applied to a suction cup, such as the suction cup **502**. That is, a surface of the suction cup that is configured to engage a surface of a component (such as a glass door of a refrigerated or freezer compartment) includes the coupling layer **603**. For example, the coupling layer **603** can coat the surface of the suction cup. As another example, the coupling layer **603** can be applied over at least a portion of the surface of the suction cup. In at least one other example, the coupling layer **603** can be applied over an entirety of the suction cup.
(90) In at least one example, a securing assembly **120**, such as any of the described herein, includes a layer of nanogel coupled to a suction cup. The securing assembly having the nanogel (such as a layer of nanogel that coats a rear surface of the suction cup) can then be mounted on a surface of a component, such as a glass panel of a door of a freezer or a refrigerated compartment. The surface of the component can first be cleaned with an alcohol wipe, for example. It has been found that the nanogel layer combats condensation that can form on freezer doors (a suction cup without such nanogel layer is less likely to combat such condensation).
(91) Further, the disclosure comprises examples according to the following clauses: Clause 1. A product information system comprising: a housing; and a communication device coupled to the housing, wherein the communication device is configured to provide information regarding one or

more products in response to being engaged by a device of an individual. Clause 2. The product information system of Clause 1, wherein the housing comprises a rear panel secured to a front panel. Clause 3. The product information system of Clauses 1 or 2, wherein the housing comprises a clip that removably retains a card, wherein the card retains the communication device. Clause 4. The product information system of Clause 3, wherein the clip comprises a tap indicia, wherein the tap indicia provides an area where the individual can tap the device to engage the communication device. Clause 5. The product information system of Clauses 3 or 4, wherein the card comprises a plurality of defined slots configured to receive and retain reciprocal protrusions of the clip. Clause 6. The product information system of any of Clauses 3-5, wherein the card comprises one or both of a graphics or text regarding the one or more products. Clause 7. The product information system of any of Clauses 3-6, wherein the communication device is secured within folded portions of the card. Clause 8. The product information system of any of Clauses 1-7, further comprising a securing assembly coupled to the housing. Clause 9. The product information system of Clause 8, wherein the securing assembly comprises a moveable joint that moveably couples the housing to a mount. Clause 10. The product information system of Clause 9, wherein the moveable joint is a ball and socket joint. Clause 11. The product information system of any of Clauses 8-10, wherein the securing assembly comprises a flexible layer of nanotechnology gel. Clause 12. The product information system of any of Clauses 8-11, wherein the securing assembly comprises a suction cup assembly. Clause 13. The product information system of Clause 12, wherein the suction cup assembly comprises a flexible layer of nanotechnology gel. Clause 14. The product information system of any of Clauses 8-13, wherein the securing assembly comprises a stand. Clause 15. The product information system of any of Clauses 8-14, wherein the communication device is secured to the securing assembly. Clause 16. The product information system of any of Clauses 8-15, wherein the securing assembly comprises a suction cup and a suction securing nut. Clause 17. The product information system of Clause 16, wherein the communication device is disposed between and within one or both of the suction cup and the suction securing nut. Clause 18. The product information system of Clauses 16 or 17, wherein the securing assembly further comprises an insert mount that fits within or around a stem of the suction cup, and wherein the communication device is mounted on the insert mount. Clause 19. The product information system of any of Clauses 16-18, wherein the communication device is secured to the suction cup. Clause 20. The product information system of Clause 16, wherein the communication device is secured to the suction securing nut. Clause 21. The product information system of any of Clauses 1-20, wherein the communication device comprises a near field communication chip. Clause 22. The product information system of any of Clauses 1-21, wherein the housing comprises a front ring removably secured to a rear ring, wherein a central passage is formed through the ring, and wherein the communication device is aligned with the central passage. Clause 23. The product information system of any of Clauses 1-22, wherein the housing comprises a front ring secured to a rear retainer, and wherein the rear retainer secures to a rail by a retaining clip and a tap screw. Clause 24. A product information method comprising: coupling a communication device to a housing; and providing, by the communication device, information regarding one or more products in response to the communication device being engaged by a device of an individual. Clause 25. A product information system comprising: a housing including a clip having a tap indicia; and a card having a communication device, wherein the clip removably retains the card, wherein the communication device is secured within folded portions of the card, wherein the card comprises a plurality of defined slots configured to receive and retain reciprocal protrusions of the clip, wherein the communication device is configured to provide information regarding one or more products in response to being engaged by a device of an individual, and wherein the tap indicia provides an area where the individual can tap the device to engage the communication device. Clause 26. The product information system of Clause 25, further comprising a securing assembly coupled to the housing. Clause 27. The product information system of Clause 26, wherein the securing assembly comprises a flexible layer of nanotechnology gel. Clause 28. The product information system of Clauses 26 or 27, wherein the securing assembly comprises a suction cup assembly. Clause 29. The product information system of Clause 28, wherein the suction cup assembly comprises a flexible layer of nanotechnology gel. Clause 30. The product information system of any of Clauses 25-29, wherein the securing assembly comprises a stand.

(92) As described herein, examples of the present disclosure provide product information systems and methods, such as can be used with a product holder system, that is configured to allow an individual to quickly, and easily determine information regarding a product, service, and/or the like.

(93) While various spatial and directional terms, such as top, bottom, lower, mid, lateral, horizontal, vertical, front and the like can be used to describe examples of the present disclosure, it is understood that such terms are merely used with respect to the orientations shown in the drawings. The orientations can be inverted, rotated, or otherwise changed, such that an upper portion is a lower portion, and vice versa, horizontal becomes vertical, and the like.

(94) As used herein, a structure, limitation, or element that is “configured to” perform a task or operation is particularly structurally formed, constructed, or adapted in a manner corresponding to the task or operation. For purposes of clarity and the avoidance of doubt, an object that is merely capable of being modified to perform the task or operation is not “configured to” perform the task or operation as used herein.

(95) It is to be understood that the above description is intended to be illustrative, and not restrictive. For example, the above-described examples (and/or aspects thereof) can be used in combination with each other. In addition, many modifications can be made to adapt a particular situation or material to the teachings of the various examples of the disclosure without departing from their scope. While the dimensions and types of materials described herein are intended to define the parameters of the various examples of the disclosure, the examples are by no means limiting and are exemplary examples. Many other examples will be apparent to those of skill in the art upon reviewing the above description. The scope of the various examples of the disclosure should, therefore, be determined with reference to the appended claims, along with the full scope of equivalents to which such claims are entitled. In the appended claims and the detailed description herein, the terms “including” and “in which” are used as the plain-English equivalents of the respective terms “comprising” and “wherein.” Moreover, the terms “first,” “second,” and “third,” etc. are used merely as labels, and are not intended to impose numerical requirements on their objects. Further, the limitations of the following claims are not written in means-plus-function format and are not intended to be interpreted based on 35 U.S.C. § 112(f), unless and until such claim limitations expressly use the phrase “means for” followed by a statement of function void of further structure.

(96) This written description uses examples to disclose the various examples of the disclosure, including the best mode, and also to enable any person skilled in the art to practice the various examples of the disclosure, including making and using any devices or systems and performing any incorporated methods. The patentable scope of the various examples of the disclosure is defined by the claims, and can include other examples that occur to those skilled in the art. Such other examples are intended to be within the scope of the claims if the examples have structural elements that do not differ from the literal language of the claims, or if the examples include equivalent structural elements with insubstantial differences from the literal language of the claims.

Claims

1. A product information system comprising: a card; a communication device retained by the card; and a housing comprising a clip that removably retains the card that retains the communication device wherein the communication device is configured to provide information regarding one or

more products in response to being engaged by a device of an individual.
2. The product information system of claim 1, wherein the housing comprises a rear panel secured to a front panel.
3. The product information system of claim 1, wherein the clip comprises a tap indicia, wherein the tap indicia provides an area where the individual can tap the device to engage the communication device.
4. The product information system of claim 1, wherein the card comprises a plurality of defined slots configured to receive and retain reciprocal protrusions of the clip.
5. The product information system of claim 1, wherein the card comprises one or both of a graphics or text regarding the one or more products.
6. The product information system of claim 1, wherein the communication device is secured within folded portions of the card.
7. The product information system of claim 6, wherein the card comprises: a base sheet having first slots; and a retainer sheet extending upwardly from the base sheet, wherein the retainer sheet has second slots, wherein the retainer sheet is folded over the base sheet, wherein the communication device is sandwiched between the retainer sheet and the base sheet, wherein the first slots are aligned with the second slots when the retainer sheet is folded over the base sheet, and wherein the first slots aligned with the second slots retain protrusions of the clip.
8. The product information system of claim 1, further comprising a securing assembly coupled to the housing.
9. The product information system of claim 8, wherein the securing assembly comprises a moveable joint that moveably couples the housing to a mount.
10. The product information system of claim 9, wherein the moveable joint is a ball and socket joint.
11. The product information system of claim 8, wherein the securing assembly comprises a flexible layer of nanotechnology gel.
12. The product information system of claim 8, wherein the securing assembly comprises a suction cup assembly.
13. The product information system of claim 12, wherein the suction cup assembly comprises a flexible layer of nanotechnology gel.
14. The product information system of claim 8, wherein the securing assembly comprises a stand.
15. The product information system of claim 8, wherein the communication device is secured to the securing assembly.
16. The product information system of claim 8, wherein the securing assembly comprises a suction cup and a suction securing nut.
17. The product information system of claim 16, wherein the communication device is disposed between and within one or both of the suction cup and the suction securing nut.
18. The product information system of claim 16, wherein the securing assembly further comprises an insert mount that fits within or around a stem of the suction cup, and wherein the communication device is mounted on the insert mount.
19. The product information system of claim 16, wherein the communication device is secured to the suction cup.
20. The product information system of claim 16, wherein the communication device is secured to the suction securing nut.
21. The product information system of claim 1, wherein the communication device comprises a near field communication chip.
22. The product information system of claim 1, wherein the housing comprises a front ring removably secured to a rear ring, wherein a central passage is formed through the ring, and wherein the communication device is aligned with the central passage.
23. The product information system of claim 1, wherein the housing comprises a front ring secured

to a rear retainer, and wherein the rear retainer secures to a rail by a retaining clip and a tap screw.
24. A product information method comprising: retaining a communication device by a card; removably retaining the card that retains the communication device by a clip of a housing; and providing, by the communication device, information regarding one or more products in response to the communication device being engaged by a device of an individual.
25. A product information system comprising: a housing including a clip having a tap indicia; and a card having a communication device, wherein the clip removably retains the card, wherein the communication device is secured within folded portions of the card, wherein the card comprises a plurality of defined slots configured to receive and retain reciprocal protrusions of the clip, wherein the communication device is configured to provide information regarding one or more products in response to being engaged by a device of an individual, and wherein the tap indicia provides an area where the individual can tap the device to engage the communication device.
26. The product information system of claim 25, further comprising a securing assembly coupled to the housing.
27. The product information system of claim 26, wherein the securing assembly comprises a flexible layer of nanotechnology gel.
28. The product information system of claim 26, wherein the securing assembly comprises a suction cup assembly.
29. The product information system of claim 28, wherein the suction cup assembly comprises a flexible layer of nanotechnology gel.
30. The product information system of claim 26, wherein the securing assembly comprises a stand.